

Kaushik S. Joshi

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SUMMARY

Machine Learning Engineer specializing in end-to-end ML pipeline design, model deployment, and automation using MLOps practices. Experienced in LLMs, NLP, and computer vision, with a strong foundation in Python, TensorFlow, and PyTorch.

EDUCATION

Master of Science (MS) in Artificial Intelligence | *Boston University*, Boston, MA *Expected: May 2027*
Coursework: AI, ML, NLP, Data Science, Multimodal ML, Image and Video Computing

Postgraduate Diploma in AI and ML | *MIT World Peace University*, India *Graduated: May 2023*

Bachelor of Science (BS) in Computer Science | *MIT World Peace University*, India *Graduated: May 2022*

TECHNICAL AND RESEARCH SKILLS

Programming – Python, SQL, Java, C++

ML & Data Science – EDA, preprocessing, feature engineering, model training, evaluation, error analysis, pandas, NumPy

Deep Learning & Vision – CNNs, image processing, object-level analysis, PyTorch, TensorFlow, OpenCV

LLMs, VLMs & RAG – Prompt design, embeddings, chunking, retrieval pipelines, multimodal benchmarks

Tools & Deployment – FastAPI, REST APIs, Docker, Git/GitHub, AWS EC2/S3, MySQL, MongoDB

Research – Literature review, experimental design, ablation studies, technical writing, teamwork, communication

WORK EXPERIENCE

High Energy Materials Research Laboratory (HEMRL) Pune, India
AI Engineer Apprentice *Aug. 2023 – May 2025*

- Contributed across the ML lifecycle, from data preparation and feature analysis to evaluation and deployment support.
- Optimized AI/CV workflows by refining model logic and reducing compute overhead by 9.4%.
- Built secure full-stack applications for local inference, supporting 15+ users in restricted environments.
- Collaborated with scientists and analysts to identify 10+ AI use cases across chemical and engineering domains.

Research Intern *Feb. 2023 – Jul. 2023*

- Researched a domain specific problem through literature review, technical gap analysis, and deep learning study.
- Proposed a novel detection approach combining image analysis and model experimentation.
- Developed an experimental UI prototype and contributed to research writing for a conference publication.

Stacklab.in Pune, India
AI/ML Engineer Intern *Jul. 2022 – Dec. 2022*

- Built and deployed ML models, integrating them with APIs and data pipelines.
- Enhanced system scalability through pipeline optimization and model retraining automation.

RESEARCH

Selecting RAG vs. File-First Deployment Strategies Across Open-Weight Reasoning Models for Efficient AI Systems

- Accepted Conference Paper – IEEE SIEDS 2026, K. S. Joshi et al.*

Unleashing the Power of Deep Learning: A Novel Approach for Defect Detection in Solid Rocket Motors

- Peer-reviewed Journal Article – FirePhysChem, A. S. Joshi, K. S. Joshi et al. (DOI: 10.1016/j.fpc.2025.06.005)*

PROJECTS

M.A.R.S. – Multi-Agent Retail Systems | *Agentic AI, Retail Automation* *May 2026*

- Designed a multi-agent retail system for inventory risk, demand forecasting, dynamic pricing, and POS/API execution.
- Automated policy-safe markdown decisions to reduce perishable waste while protecting store-level profit margins.
- Coordinated inventory, pricing, risk, and execution agents to support autonomous retail decision-making.

Spatial Reasoning Benchmark for Multimodal VLMs | *Multimodal AI, VLMs, Interpretability* *May 2026*

- Built a unified evaluation framework for VLM spatial reasoning across relation, viewpoint, counting, and grounding tasks.
- Benchmarked Qwen2.5-VL, Cambrian, Phi-3.5-Vision, BLIP-2, and InstructBLIP across diverse spatial datasets.
- Analyzed failures using benchmark difficulty, model agreement, Grad-CAM, attention patterns, and generalization gaps.

RARC – Reasoning-Aware Retrieval Controller | *LLMs, RAG, Retrieval Evaluation* *April 2026*

- Developed a novel adaptive RAG approach that routes queries and adjusts retrieval depth, context size, and answer mode at inference time.
- Evaluated RARC against Fixed RAG and no-retrieval LLM baselines, reducing hallucination and improving answer relevance across multi-model experiments.

ACHIEVEMENTS AND EXTRACURRICULAR

- Won 3rd Prize at Boston University Med AI Hackathon. *Apr. 2026*
- Secured 3rd Place in CODEMENT_24, a national-level 24-hour hackathon. *Apr. 2024*
- Awarded Best Volunteer for 4 years of service with MIT NCC Troop. *Apr. 2023*
- Received Most Creative Award in BS Computer Science. *May. 2022*